

CANDIDATE'S NAME: _____

CTG: _____

YISHUN JUNIOR COLLEGE
2018 JC2 PRELIMINARY EXAMINATION

CHEMISTRY
HIGHER 2

9729/01

Paper 1 Multiple Choice

THURSDAY 13 SEPTEMBER 2018

1400hrs – 1500hrs

(1 hour)

Additional Materials: Optical Mark Sheet (OMS)
Data Booklet



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, CTG, and NRIC / FIN number on the Optical Mark Sheet (OMS), and shade the corresponding boxes for your NRIC / FIN number.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

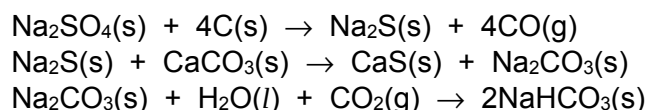
The use of an approved scientific calculator is expected, where appropriate.

- 1 10 cm³ of a pure hydrocarbon was completely burned in 80 cm³ of excess oxygen at 425 K. After cooling to room temperature, the volume of the gaseous mixture was found to be 55 cm³. A reduction of 40 cm³ was observed when the residual gases were passed through calcium hydroxide. All gas volumes were measured at the same temperature and pressure.

What is the formula of the hydrocarbon?

- A** C₄H₁₀ **B** C₄H₈ **C** C₂H₆ **D** C₂H₄

- 2 Sodium hydrogen carbonate, NaHCO₃, can be prepared from sodium sulfate by a three-step process as shown.



Assuming a yield of 90% in each step, determine the mass of sodium hydrogen carbonate, to the nearest kg, that could be obtained from 100 kg of sodium sulfate.

- A** 146 kg **B** 122 kg **C** 86 kg **D** 43 kg

- 3 Ions of the two common isotopes of nickel are shown below.

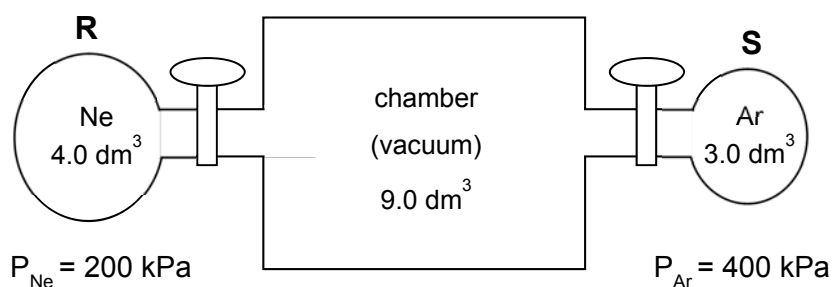


Which statements about these ions are correct?

- 1 ⁵⁸Ni²⁺ ion has a greater number of nucleons in its nucleus than the ⁶⁰Ni²⁺ ion.
- 2 Both ions have the same number of protons.
- 3 Both ions have more neutrons than electrons.
- 4 A beam of ⁵⁸Ni²⁺ ion is deflected to a larger extent than that of ⁶⁰Ni²⁺ ion in an electric field.

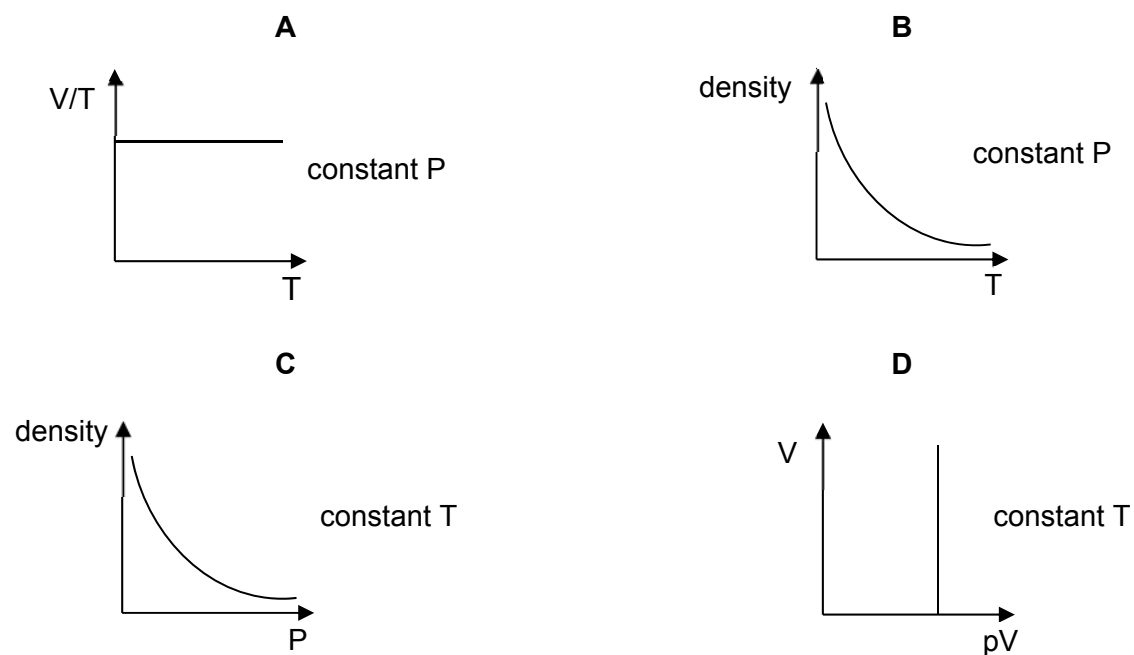
- A** 1, 2, 3 and 4
B 1, 2 and 3
C 2 and 3
D 2, 3 and 4

- 4 Two bulbs **R** and **S** are connected to a 9 dm^3 vacuum chamber as shown.

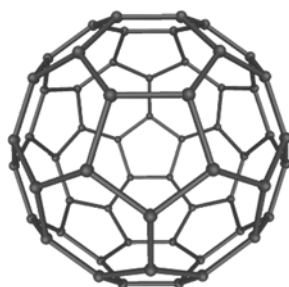


What will be the total pressure in the vessel when the valves are opened at constant temperature?

- A** 600 kPa
B 286 kPa
C 222 kPa
D 125 kPa
- 5 Which graph incorrectly describes the behaviour of a fixed mass of an ideal gas?



- 6 Allotropes of carbon include diamond, graphite and fullerene. A spherical fullerene, known as buckminsterfullerene, has the formula C_{60} with the following structure.



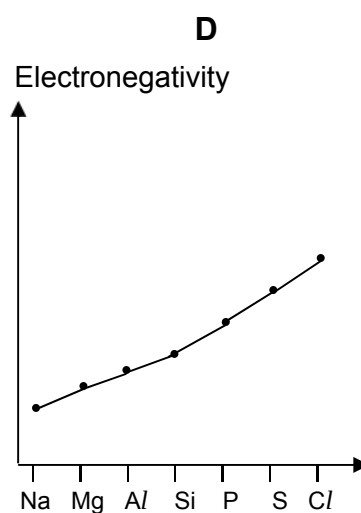
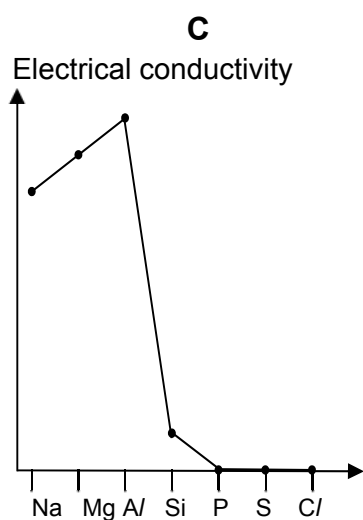
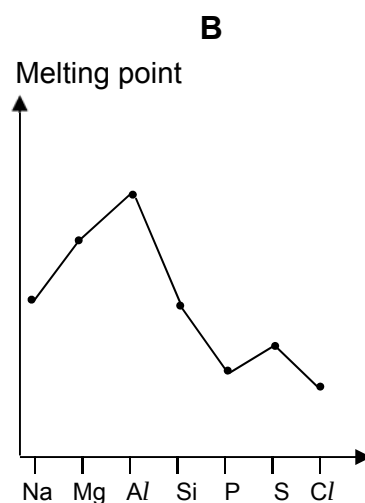
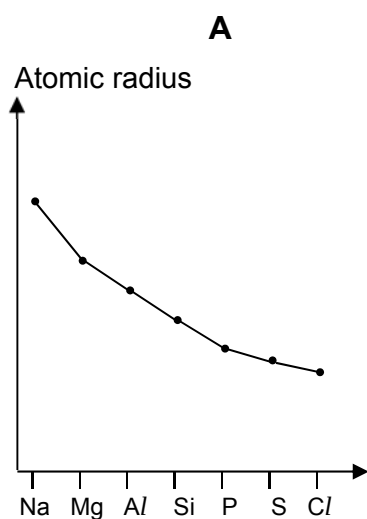
Which statement is correct?

- A All the bond angles in buckminsterfullerene are 120° .
 - B There are delocalised electrons within buckminsterfullerene.
 - C Buckminsterfullerene sublimes at a higher temperature than graphite.
 - D On complete combustion, buckminsterfullerene forms carbon dioxide and water.
- 7 The enthalpy change of solution of magnesium fluoride is found to be negative while that of barium fluoride is positive.

Which statement best explains this observation?

- A Magnesium fluoride is soluble in water, but barium fluoride is insoluble.
- B The hydration energy of the barium ion is bigger in magnitude than that of the magnesium ion.
- C The lattice energy of barium fluoride is bigger in magnitude than that of magnesium fluoride.
- D The lattice energy of barium fluoride is bigger in magnitude than the sum of the hydration energies of the barium and fluoride ions.

- 8 Which sketch shows the incorrect trend for the stated property for elements in the third period of the Periodic Table?



- 9 The table shows the results of experiments in which the halogens X_2 , Y_2 and Z_2 are added to aqueous solutions containing X^- , Y^- , and Z^- ions.

	$X^-(aq)$	$Y^-(aq)$	$Z^-(aq)$
X_2	no reaction	no reaction	no reaction
Y_2	X_2 formed	no reaction	Z_2 formed
Z_2	X_2 formed	no reaction	no reaction

Which row shows the correct order of the reducing strength of the halide ions?

- | | | | |
|----------|-----------|-------|---------|
| | strongest | | weakest |
| A | X^- | Z^- | Y^- |
| B | Y^- | Z^- | X^- |
| C | Z^- | Y^- | X^- |
| D | X^- | Y^- | Z^- |

10 The diagram below shows the positions of four elements **W**, **X**, **Y** and **Z** in periods 2 and 3.

	W		
X		Y	Z

Element **Y** has the following successive ionisation energies(IE).

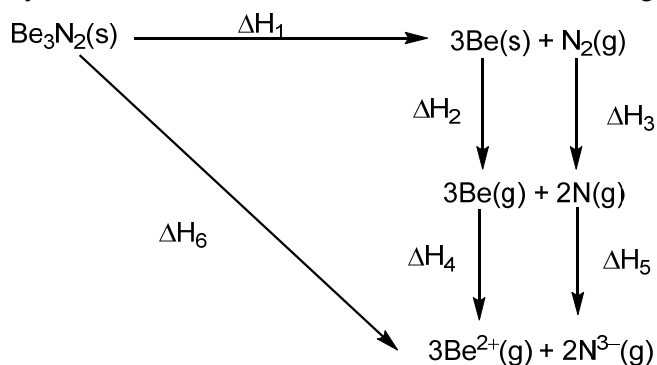
IE /kJ mol ⁻¹	1011	1907	2914	4963	6273	21268	29872
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Which statement about elements **W**, **X**, **Y** and **Z** and their compounds is incorrect?

- A** The oxide of **X** is insoluble in water.
- B** The first ionisation energy of **Z** is lower than that of **Y**.
- C** When HCl(aq) is added to a sample containing oxide of **X**, the solid dissolves.
- D** **W**, **Y** and **Z** form chlorides which will react with water to produce acidic solutions.

11 Use of Data Booklet is relevant to this question.

The lattice energy of beryllium nitride can be determined via the following energy cycle.



Enthalpy Change	$\Delta H / \text{kJ mol}^{-1}$
ΔH_2	+324
ΔH_5	+1404
ΔH_6	+11300

Which statements are correct?

- The enthalpy change of formation, ΔH_f , of Be_3N_2 is +648 kJ mol⁻¹.
 - ΔH_3 is always endothermic.
 - The average energy of the electrostatic forces of attraction present in Be_3N_2 is 11300 kJ mol⁻¹.
- A** 1 only
 - B** 2 and 3 only
 - C** 1 and 2 only
 - D** 1, 2 and 3

12 In which processes will ΔS be negative?

- 1 The burning of sulfur in oxygen to produce sulfur dioxide gas.
- 2 The dissolution of solute in solvent.
- 3 The synthesis of an ionic compound from its constituent gaseous ions.

- A** 3 only
B 1 and 3 only
C 1 and 2 only
D 1, 2 and 3

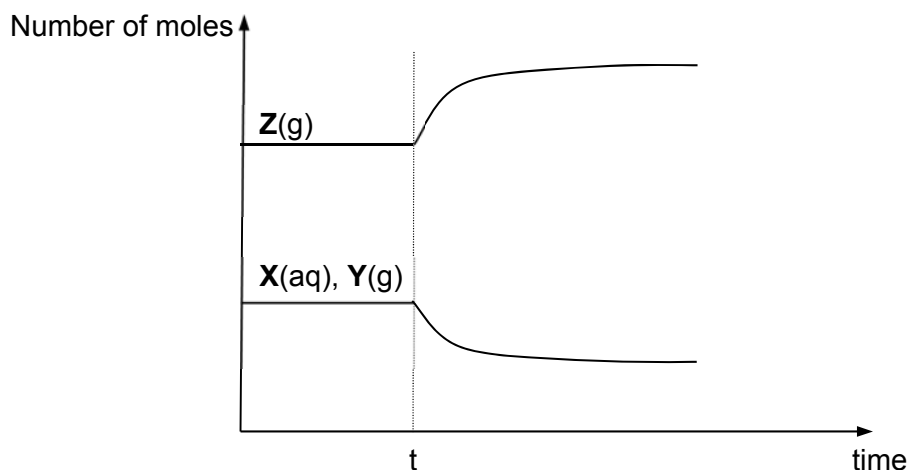
13 A bomb calorimeter contains 225 dm^3 of hydrazine, N_2H_4 as fuel. After combustion, the temperature change in the calorimeter was found to be $w^\circ\text{C}$.

The heat capacity of the calorimeter is $z \text{ kJ K}^{-1}$. The density of liquid hydrazine is 1.02 g cm^{-3} and its enthalpy change of combustion, ΔH_c , is $-622.2 \text{ kJ mol}^{-1}$

Which of the following represents the percentage efficiency of the bomb calorimeter?

- A** Percentage efficiency = $\frac{225 \times 1000 \times 1.02 \times z \times w}{622.2} \times 100\%$
- B** Percentage efficiency = $\frac{225 \times 1000 \times 1.02 \times z \times (w + 273)}{622.2} \times 100\%$
- C** Percentage efficiency = $\frac{(w + 273) \times z}{\frac{225 \times 1000 \times 1.02}{\text{Molar mass (N}_2\text{H}_4)}} \times 622.2} \times 100\%$
- D** Percentage efficiency = $\frac{w \times z}{\frac{225 \times 1000 \times 1.02}{\text{Molar mass (N}_2\text{H}_4)}} \times 622.2} \times 100\%$

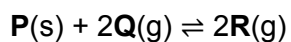
- 14 Consider the following reaction: $\text{X(aq)} + \text{Y(g)} \rightleftharpoons \text{Z(g)}$ $\Delta H < 0$
The graph shows the number of moles of gases in a reaction mixture as it changes with time.



The reaction mixture is initially at equilibrium until time, t , when a disturbance is introduced to the system. Which of the disturbances at time, t , can account for the observed change in the graph above?

- 1 An increase in pressure
 - 2 Addition of water into the aqueous solution
 - 3 Increase in temperature
- A** 1 only
B 2 only
C 1, 2 and 3
D None of the above

- 15 Given that the K_p for the following equilibrium is 64,

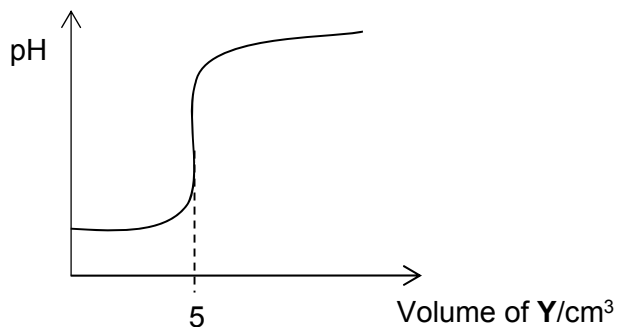


What is the mole ratio of **Q** : **R** at equilibrium?

- A** 1 : 1
B 1 : 8
C 8 : 1
D 64 : 1

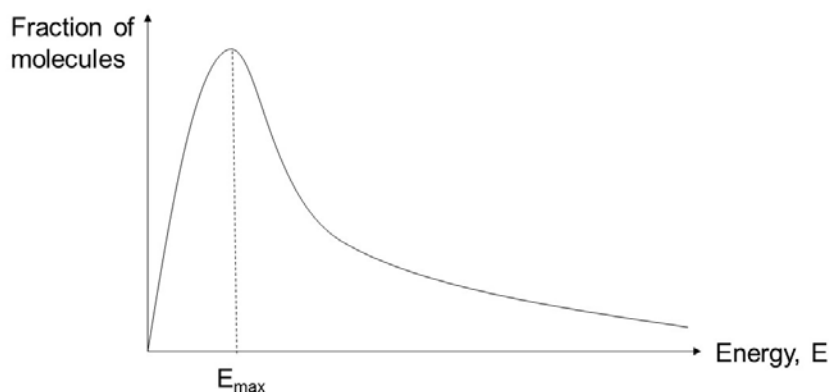
- 16 The graph below shows how the pH changes when 0.1 mol dm^{-3} solution of **Y**(aq) was gradually added to 10 cm^3 of 0.1 mol dm^{-3} solution of **X**(aq).

What could **X** and **Y** be?



	X	Y
A	NaOH	HCl
B	HCl	NaOH + KOH
C	HCl	Na_2CO_3
D	HCl	NaHCO_3

- 17 The diagram below represents the Boltzmann distribution of molecular energies at a given temperature.



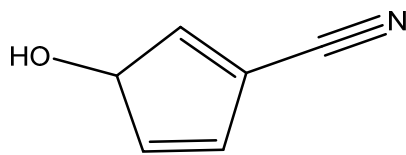
With an increase in temperature, which statements are correct?

- 1 At all energies, the proportion of molecules increases.
- 2 The peak of the graph decreases.
- 3 The value of the rate constant, k , increases.

- A** 1 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

18 Use of Data Booklet is relevant to this question.

Solid compound **J** has the following structure. It undergoes complete combustion in air to produce carbon dioxide, water and nitrogen dioxide only.

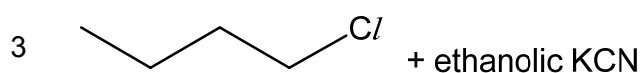
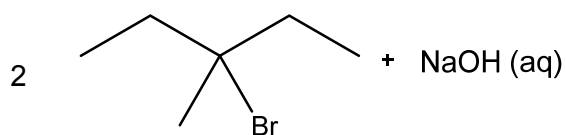
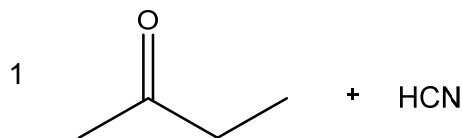


Compound **J**

Which statement about compound **J** is correct?

- A** On complete combustion, 0.10 g of compound **J** produces 0.230 g of CO_2 .
- B** On complete combustion, 0.10 g of compound **J** produces 22.4 cm^3 of NO_2 measured under room conditions.
- C** 0.10 g of compound **J** reacts with excess bromine in organic solvent to produce 0.380 g of product.
- D** 0.10 g of compound **J** reacts with hydrogen gas and platinum catalyst to produce 0.100 g of product.

19 Which reaction will produce a mixture which is optically inactive?

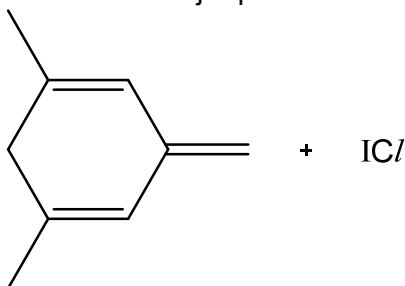


- A** 1, 2 and 3
- B** 1 and 2 only
- C** 1 only
- D** 2 and 3 only

20 How many constitutional isomers are possible for a substituted benzene, C_8H_{10} ?

- A 3
- B 4
- C 5
- D 6

21 How many chiral carbons are formed in the major product of the following reaction?



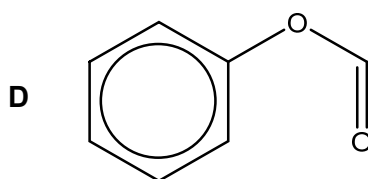
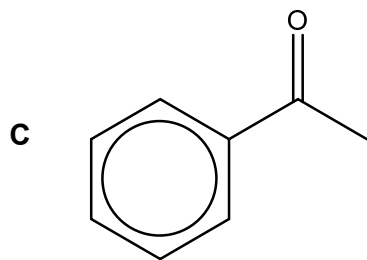
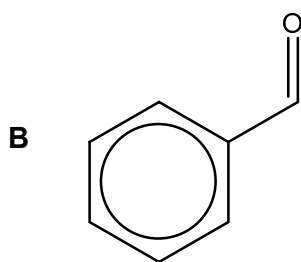
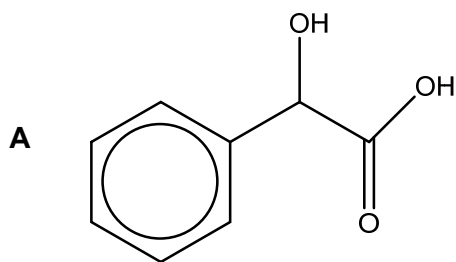
- A 2
- B 3
- C 4
- D 5

22 Which statement best explains why benzene undergoes substitution reaction while ethene undergoes addition reaction?

- A Benzene provides steric hindrance to repel reactive species.
- B The C=C bonds in benzene are shorter and stronger than the C=C bonds in ethene.
- C There is delocalisation of π electrons in benzene which is absent in ethene.
- D Benzene contains more C=C bonds than ethene, hence it is more electron rich.

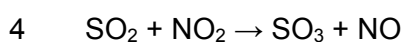
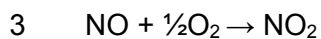
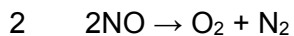
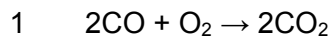
- 23** Benzoic acid can be obtained using either a one-step or two-step synthesis.

Which of the following cannot be used as the starting material?



- 24** A catalytic converter is a device used to reduce the toxicity of emissions from an internal combustion engine. It works by using a catalyst (platinum, palladium or rhodium) to stimulate a chemical reaction in which toxic by-products of combustion are converted to less toxic substances.

Identify the chemical reactions occurring.



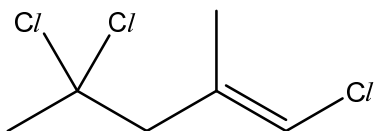
A 1 and 2 only

B 2 and 3 only

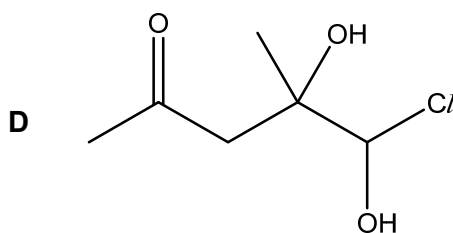
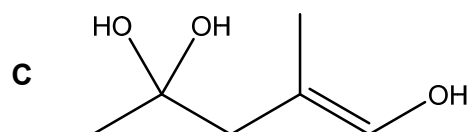
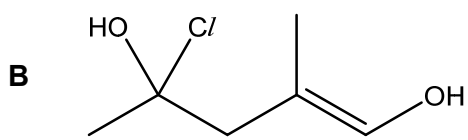
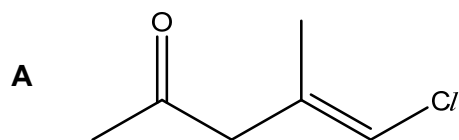
C 1, 2 and 3 only

D 3 and 4 only

- 25 The following compound is heated with dilute sodium hydroxide.



Which of the following product is obtained?



- 26 Consider the following four compounds.

- 1 propanol
- 2 phenol
- 3 propanoic acid
- 4 water

What is the relative order of **decreasing** acidity of these compounds?

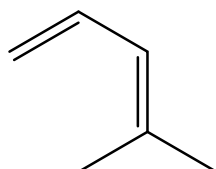
- A** 1 2 4 3
B 3 2 1 4
C 3 2 4 1
D 1 4 2 3

- 27 Compound **X** decolourises aqueous bromine. Compound **X** reacts with hot acidified potassium manganate(VII) to form compound **Y**. The following table shows some tests and observations carried out on compound **Y**.

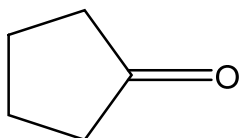
compound	Tests	Observations
Y	Na_2CO_3	Effervescence observed. Gas evolved forms white precipitate with limewater
	2,4-dinitrophenylhydrazine	Orange precipitate

What could compound **X** be?

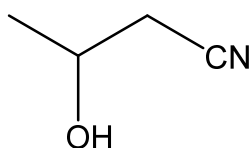
A



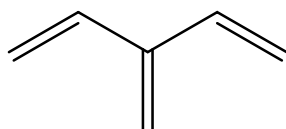
B



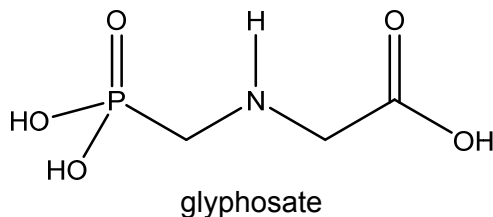
C



D

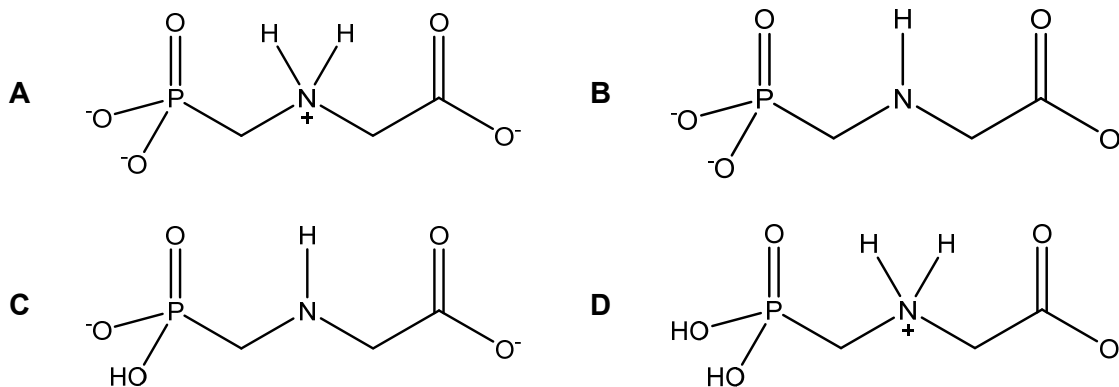


- 28 Glyphosate is a herbicide used to kill weeds.



Glyphosate has pK_a values of 0.8, 2.3, 6.0 and 11.0.

What is the structure of the major species in a solution of glyphosate at pH 7?

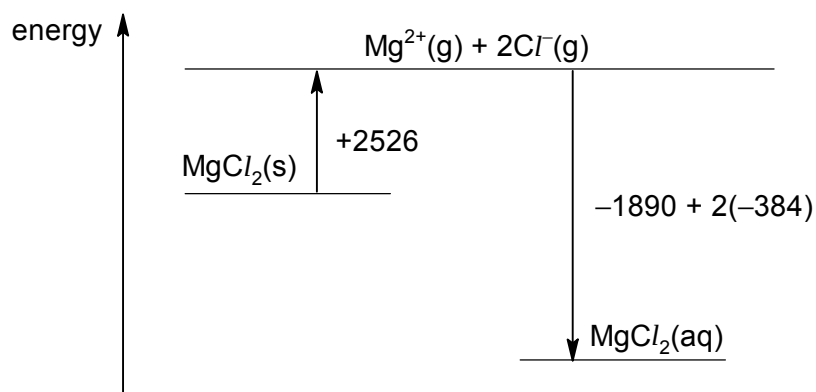


- 29 *Use of Data Booklet is relevant to this question.*

Which of the following reactions of transition metals is incorrect?

- A** Addition of $\text{H}_2\text{O}_2(\text{aq})$ to acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ produces a green solution.
- B** V^{3+} is a suitable catalyst for the reaction: $2\text{I}^- + \text{S}_2\text{O}_8^{2-} \rightarrow \text{I}_2 + 2\text{SO}_4^{2-}$.
- C** Addition of concentrated HCl to $\text{CuSO}_4(\text{aq})$ produces a yellow solution.
- D** Addition of $\text{Na}_2\text{CO}_3(\text{aq})$ to $\text{Cr}^{3+}(\text{aq})$ produces a gas that gives a white precipitate when bubbled through limewater.

- 30 The following energy level diagram represents the dissolving of anhydrous magnesium chloride in a large volume of water.



Which statements about the process are correct?

- 1 The lattice energy of magnesium chloride is $-2526 \text{ kJ mol}^{-1}$.
 - 2 The enthalpy change of hydration of the chloride ion is -384 kJ mol^{-1} .
 - 3 The enthalpy change of solution of anhydrous magnesium chloride is -132 kJ mol^{-1} .
 - 4 $\text{Mg}^{2+}(\text{g})$ and $\text{Cl}^{-}(\text{g})$ are less thermally stable than $\text{MgCl}_2(\text{aq})$.
- A** 1, 2, 3 and 4
- B** 1, 2 and 3 only
- C** 2 and 3 only
- D** 2 and 4 only